

Go for it, kick, or punt?

- A fourth-down decision engine built on every NFL play since 2014
- Version 1.2.0

Fourth down is a choice between three gambles

- Go for it: keep the drive, or hand the other team a short field
- Field goal: three likely points, but you give up the ball
- Punt: give up the ball, but pin them deep

We ask one question

- After each choice, what is our chance to win the game?
- Points alone miss the score and the clock: a tying kick with 4 seconds left is worth far more than 3 points
- The choice with the best chance to win is the call

What it learned from

- 580,005 plays from 2014 to 2025
- How often teams win from every score, clock and spot on the field
- How often teams convert, how often kicks go in, where punts land
- How much time each kind of play takes off the clock

Everything that matters goes in

- Always: score, time left, yards to go, field position, both teams' timeouts, the two-minute warning, who gets the ball at halftime
- Switch on or off: wind, cold and rain · dome and altitude · how good each team is · your actual kicker

How sure is it?

- We re-built the whole engine 20 times on reshuffled games
- If almost all 20 agree, the call is confident
- If they split, it is a toss-up, and neither choice is a mistake

Does it work?

- We tested it on two seasons it never saw
- When coaches went for it, it predicted a 37% chance to win; those teams won 38%
- Field goals: predicted 56%, actual 56% · Punts: predicted 47%, actual 47%

Every coach's call, played twice

- Once the way the coach called it, once the way the engine would
- A coach is only charged when the engine is sure he was wrong
- Coaches went for it on 13% of fourth downs in 2014 and 24% in 2025
- Closest to the math: Liam Coen, Sean McDermott, Matt LaFleur
- Furthest from it: Shane Steichen, Todd Bowles, Sean Payton
- Trust the order more than the exact numbers: the engine's totals run higher than published research

Kickers matter

- A replacement-level kicker makes 61% from 50 yards; an average one makes 77%
- The engine can use your actual kicker's accuracy and range
- Best current kickers above replacement: B.Aubrey, C.Boswell, W.Reichard

Built for game day

- Fill from a live game: pick the game and your side, and the score, clock, down and spot fill in
- Pick the two teams and the stadium, roof and your kicker are set for you
- Share any call as a link that opens straight to the answer
- Every Tuesday: the week's costliest calls and best gutsy ones
- willhoop.github.io/fourth-down — add it to your phone's home screen

What it can't tell you

- '4th and 1' can mean inches or a full yard; the data can't tell them apart
- It can't prove that following it wins more games: no one has run that experiment
- Small edges are toss-ups, not mistakes

Read the full story

- White paper: method, math, results and sources
- docs/fourth-down-whitepaper.md · build/fourth-down-whitepaper.pdf